

FARMCOM CORE AGRI-INTELLIGENCE SYSTEMS

Algorithmic Aggregation Framework, Offline-First Yield Logistics Data Networks, and International Trade Compliance Infrastructure

Strategic Alignment: Track 6 (Agriculture and Agribusiness Management) & Track 12 (Monitoring & Evaluation Systems) | Target Horizon: 2026 – 2031 | Operational Framework

1. Executive Summary & Market Mechanics

Agriculture generates over 24% of Uganda's Gross Domestic Product, yet the underlying commercial market is held back by fragmented smallholder supply lines. Institutional buyers, commodity processors, and international trade off-takers face major obstacles due to volatile regional pricing and highly unpredictable harvest volume forecasting. Concurrently, strict global compliance frameworks—such as the European Union Deforestation Regulation (EUDR)—require absolute proof that agricultural exports are completely disconnected from deforested land patches, creating a critical market access risk for local coffee and cocoa exporters.

FarmCom Core addresses this structural supply chain friction by deploying an offline-first data aggregation platform managed by cooperative coordinators and enterprise field teams. Moving far past low-retention consumer chat boards, FarmCom Core digitizes crop conditions and agronomic indicators at the individual farm plot level, organizing unstructured regional yields into large, verifiable batches for international corporate buyers.

2. Technical Core & Intelligent Systems

Mobile Architecture

Cross-platform field applications engineered using Flutter and Dart, utilizing a local reactive database engine (SQLite/WatermelonDB) to ensure seamless data logging in deep rural zones lacking cellular connectivity.

Synchronization Gateway

Asynchronous background service workers that queue local modifications on the device and securely transmit data payloads to a centralized FastAPI infrastructure the moment a stable network handshake is established.

Geospatial Layer

Leverages PostgreSQL supercharged with PostGIS database extensions to process, store, and query complex farm boundary polygons mapped on-site by field teams.

Algorithmic Predictive Yield Modeling

To provide bulk commercial commodity buyers with dependable volume projections weeks before harvest, the core engine runs localized field metrics through an ensemble regression architecture:

$$Y_{pred} = \Sigma (A_i \times \eta_i \times \Omega_m)$$

Where Y_{pred} represents the calculated crop volume forecast for a targeted cooperative zone, A_i is the mapped physical acreage of farm plot i , η_i is the plant density index, and Ω_m is a dynamic environmental coefficient computed by cross-referencing satellite canopy indices with localized historical weather trends.

3. Market Strategy & Monetization Matrix

Revenue Vector	Operational Model	Target Consumer Tier
B2B Clearing Commission	A transactional brokerage percentage fee applied directly to bulk purchasing contracts cleared and settled through the platform.	Large Agro-Processors & Global Exporters
Data Intelligence SaaS	Premium subscription tiers granting enterprise access to regional yield tracking dashboards and supply risk metrics.	Microfinance Institutions & Commodity Lenders
Traceability Verification	Flat pricing tier applied per batch ledger generated to	Coffee, Cocoa, and

	provide unalterable compliance documentation for international customs.	Timber Export Firms
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4. Market Sizing (estimates)

TAM (Global agri-traceability & intelligence software): \$2.5B.

SAM (East Africa specialty export market): \$250M.

SOM (3-year target capture): \$4.5M via cooperatives and exporters in targeted corridors.

5. Unit Economics & Pricing

Enterprise dashboard subscriptions: \$5k–\$25k/year depending on region and data scope. Marketplace commission typical 3–7% per transaction. Expected gross margin on SaaS and data: 70–85%. CAC estimate for cooperative onboarding: ~\$2k; expected LTV: ~\$40k over 5 years.

6. Financial Projections (3-year summary)

Year 1 Revenue: \$0.6M

Year 2 Revenue: \$2.0M

Year 3 Revenue: \$6.0M

Target EBITDA margin by Year 3: ~40%.

7. Go-to-Market

Channels: partnerships with exporters, cooperatives, NGOs and development agencies; subsidized pilots co-funded by donors to de-risk initial deployments; direct enterprise sales to commodity buyers.

8. Team & Hiring (initial)

Founders: Product lead (ag-tech), Head of Partnerships (export industry), ML lead (yield forecasting). Early hires: field ops managers, mobile engineers, and partnership managers.

9. Ask & Use of Funds (example seed)

Use	Amount (USD)
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Product & mobile ops	\$700,000
Partnership & pilots	\$350,000
GTM & scaling	\$300,000

10. Risks & Mitigations

Adoption risk: cooperative inertia and device access. Mitigation: subsidized device bundles, local field agent programs and NGO partnerships.